

Geometric Dynamic Equivalence Method for Digital Twin-Driven Embodied Agents Learning

Xin Li, Shiyuan Zhang, Yanmin Zhou, Ping Lu, Zhipeng Wang, Rong Jiang, Hongrui Sang and Bin He

Abstract— Digital Twin (DT) technology, which enables real-time spatiotemporal mapping of physical objects, offers a promising way to provide embodied agents with dynamic, reality-mirrored virtual learning environment. However, most existing agent-DT systems either rely on heavy 3D point-cloud reconstruction or on loosely coupled geometric-parameter synchronization, making it difficult to efficiently maintain geometric dynamic equivalence between complex indoor scenes and their virtual counterparts. This paper introduces *VirEqPhy*, a geometric dynamic equivalence method that formulates virtual-physical consistency as a voxel-based multi-task cascade that jointly perform CAD model retrieval and 9-DoF pose estimation from a single RGB image. *VirEqPhy* embeds observed objects and CAD models into a shared voxel feature space, couples a two-stage rotation estimation scheme with Procrustes-based refinement, and optionally integrates a weakly supervised Top-k retrieval module that learns object-CAD associations from multi-similarity labels without dense manual correspondences. In the supervised setting, *VirEqPhy* achieves 24.2% high-threshold mean AP for model retrieval and 24.4% class-level alignment accuracy under the Scan2CAD criterion. In the weakly supervised setting, *VirEqPhy* achieves 29.2% high-threshold mean AP and maintains strong performance on unseen categories. In both settings, it surpasses representative baselines. A real-robot indoor navigation experiment further verifies its effectiveness in dynamic real-world environments. For additional details, please refer <https://forbiddenname.github.io/VirEqPhyDT>.

Index Terms— Digital twin, embodied agents, geometric dynamic equivalence, model retrieval and alignment.

I. INTRODUCTION

Robust deployment of agents in indoor living scenarios for embodied tasks—such as interactive navigation [1] and object manipulation [2]—is a primary focus in embodied artificial intelligence. With rapid advances in computer graphics and high-performance physics simulation engines, robot simulators such as iGibson [3], AI-Habitat [4], and AI2-THOR [5] provide high-fidelity virtual spaces that closely mimic real-world environments. These platforms allow agents to learn diverse tasks at low cost and with high iteration rates.

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However, unlike structured settings, (e.g., manufacturing workshops), indoor living environments exhibit unpredictable geometric dynamics involving object additions, removals, and pose changes. To enable interactive learning for agents, the virtual space therefore needs to maintain geometric dynamic equivalence with the physical space, i.e., to track real-time geometric changes in the physical environment [1]. Most existing simulators, however, rely on static, well-defined object representations and lack real-time physical-virtual synchronization [6]. Driven by the concept of the metaverse, digital twin (DT), characterized by real-time mapping, has emerged as an effective bridge between the physical and virtual worlds, offering a promising method for monitoring and optimizing complex physical systems [7-9]. DT-driven virtual environments reflect real-time spatiotemporal dependencies among physical objects [10-16], providing a methodological foundation for addressing the above problem.

Existing virtual-space updating methods for agent-DT systems can be broadly categorized into 3D point cloud-based methods [12-14] and geometric parameter-based methods [6-9]. The former reconstruct object surfaces from real-time environmental point cloud data, enabling rapid responses to scene changes, but requiring computationally intensive data processing and registration algorithms for point clouds. The latter instead synchronize the geometric parameters of offline-constructed 3D models via 2D object detection, pose estimation, and model correction. Although easier to deploy, their accuracy depends critically on object detection performance and environmental conditions (e.g., lighting). As indoor agent applications grow in complexity, objects are often arranged in cluttered, occluded configurations that prevent agents from acquiring complete geometric information (e.g., point clouds or bounding boxes of occluded objects). Under these conditions, both types of methods face significant limitations. Moreover, more challenging scenarios arise when unseen objects are introduced. In such cases, point cloud-based methods cannot generate meaningful 3D models, while parameter-based methods fail due to the absence of corresponding 3D models.

Advances in computer graphics have facilitated the development of 3D model libraries (e.g., CAD and mesh models) that provide precise geometric representations of diverse objects. Building on this, model retrieval and alignment pipelines—combining 2D object detection, 3D model retrieval, and spatial alignment—offer a promising way to address the above limitations [17-24]. These approaches retrieve the 3D model of a target object from a database using

a single-perspective RGB scene image and then spatially align the model to update the virtual space. By leveraging visible features (e.g., edges, textures) instead of explicitly reconstructing geometry, they can infer complete shapes despite partial occlusion. Fusing 2D and 3D information further enables mapping of previously unseen objects to the most similar library models, supporting conditional scene updates. Nevertheless, existing methods still exhibit several limitations. *Synthesis-analysis methods* (e.g., IM2CAD [17], Total3D [18]) infer full 3D scenes but rely on simplified layout assumptions. *Two-stage pipelines* (e.g., Mask2CAD [19] and Patch2CAD [20]) offer high inference efficiency yet suffer from scale ambiguity. *End-to-end joint reasoning methods* (e.g., ROCA [21], FastCAD [22]) enhance robustness and accuracy via joint optimization of model retrieval and pose estimation but struggle with unseen objects. *Weakly supervised models* (e.g., DiffCAD [23], InCAD [24]) improve practicality in data-scarce settings, yet slow diffusion sampling and dependence on synthetic data limit their suitability for real-time DT. How to mitigate cross-domain distribution gaps under limited agent viewpoints and to establish stable 2D-to-3D mappings therefore remains a challenge.

Motivated by these gaps, we propose ***VirEqPhy***, an end-to-end virtual-physical space equivalence approach to support DT-driven embodied agent learning. ***VirEqPhy*** enables real-time tracking of geometric changes in complex real-world indoor scenarios and dynamically updates CAD model-based virtual representations. Specifically, **first**, instead of relying on point-cloud reconstruction or geometric-parameter synchronization in existing digital twin system [6–9] [12–14], ***VirEqPhy*** adopts a lightweight model retrieval-and-alignment pipeline. By embedding observed objects and CAD models into a shared, voxel-based feature space, it infers object geometry directly from appearance features and thus avoids explicit point-cloud reconstruction. This design also reduces dependence on precise semantic labels and enables robust retrieval even for occluded objects. **Second**, by coupling complete 3D voxel representations with a differentiable Top-k retriever and a learnable weighted two-stage rotation estimation strategy, ***VirEqPhy*** alleviates the reliance of prior retrieval and alignment methods on 2D cues and their sensitivity to scale and pose ambiguities [18–24], thereby improving robustness in cluttered scenes with occlusions and unseen objects. In summary, the key contributions of this work are as follows.

1) We propose a novel multi-task cascade learning framework for virtual-physical space equivalence, facilitating robust deployment of embodied agents in complex indoor scenarios. The framework uses 3D voxels as a cross-modal medium between images and CAD models to mitigate occlusion-induced feature degradation. By integrating cascaded neural networks with multi-task learning, it captures inter-task correlations and enables simultaneous execution of multiple tasks (e.g., feature extraction, voxel representation learning, model retrieval, and alignment). 2) To improve the framework’s reliability under limited matching samples, we introduce a plug-and-play weakly supervised module that

constructs a fuzzy mapping mechanism between 2D images and 3D CAD models using appearance, semantic, and geometric similarities. A differentiable Top-k algorithm with positive and negative samples is designed to optimize the joint embedding space at low cost, improving retrieval robustness for unseen objects and reducing dependence on object-cad associations. 3) We conduct extensive qualitative and quantitative experiments on supervised datasets (i.e., ScanNet-ShapeNet alignment), self-constructed weakly supervised datasets (more than 20,000 instances), and embodied agent systems in both simulation platforms and the real world (e.g., indoor navigation robots). The results demonstrate the effectiveness of the proposed method in achieving virtual-physical geometric dynamic equivalence and robust cross-domain generalization, as well as its feasibility for seamless integration into real embodied agent systems.

The rest of this paper is organized as follows. Section II reviews related work, and Section III outlines the proposed method. Sections IV and V detail the multi-task cascade learning framework and the weakly supervised module. Section VI presents experiments and results, followed by a summary and outlook in Section VII.

II. RELATED WORK

To contextualize our approach, we review related work across three domains. Section II-A traces the evolution from virtual simulators to DT for embodied agents. Section II-B reviews geometric updating strategies in DT systems, highlighting the need for a more efficient dynamic equivalence framework. Section II-C surveys model retrieval and alignment pipelines that can be used for DT updating and analyzes the open issues that directly motivate our method.

A. From Virtual Simulators to Digital Twin for Agents

With the integration of real-world scene scanning and physical simulation engines, an increasing number of advanced robotic simulation platforms provide interactive, controllable, and reusable virtual environments for agents to develop perception-decision-action algorithms [25]. Specifically, general platforms like Gazebo and Isaac Sim [26] support diverse embodied tasks but demand substantial computational resources. Specialized platforms such as iGibson [3] and AI-Habitat [4] offer high-fidelity home environment simulations. iGibson focuses on complex, long-term tasks such as room organization and object search, while AI-Habitat is well suited to navigation and human-robot interaction tasks. Physics-focused platforms like MuJoCo and PyBullet [27] excel in tasks requiring precise physical interaction (e.g., peg-in-hole assembly). However, static virtual environments lack real-time geometric synchronization with physical spaces. Consequently, algorithms trained in these platforms transfer poorly to dynamic real-world scenarios where object positions and behaviors change unpredictably—such as the addition of obstacles or the movement of target objects during interactive navigation.

To address this, some studies integrate digital twin technology into the construction of virtual environments for

agent learning [6-11, 14,15]. DT refers to the creation of real-time virtual representations of physical systems using digital means, originally developed for product lifecycle management and now widely applied to the monitoring and optimization of complex physical systems [7]. This paradigm enhances geometric realism, enabling robust agent deployment in complex embodied tasks. A DT framework with an efficient update response mechanism has been designed to enhance the robustness of navigation robot deployment in real environments [6]. A DT-assisted skill learning framework has been applied to complex 3C assembly tasks to generate feasible policy sequences [7]. An LLMs driven, multielement decision-making framework based on DT has been presented for human-robot collaborative manipulation [8]. Other applications include task replanning of robotic tasks under uncertain conditions [10], robot assembly monitoring [9, 11], human-robot safe collaboration [14], and adaptive task allocation in robotic construction [15]. These studies show that DT can reconstruct virtual spaces with both visual and physical fidelity and supports a wide range of embodied tasks. Critically, DT technology facilitates real-time geometric realism in such virtual environments, enabling robust agent deployment in complex real-world settings. However, enabling such dynamic virtual spaces also poses a key challenge: how to efficiently update the virtual counterpart in response to the physical system changes [28].

B. Geometric Updating in Digital Twin

In robotic DT systems, 3D point cloud reconstruction can theoretically enable virtual space updates applicable to arbitrary scenarios. This approach collects the 3D point clouds of real objects via depth sensors, transmits them to the virtual space, and forms a complete 3D model through registration algorithms. The key challenges are efficient point clouds transmission and registration. An iterative point clouds registration method has been proposed to accurately update DTs of equipment management [12]. A general AI-powered transmission framework has been introduced to enhance the update efficiency [13]. A deep learning-based point cloud instance segmentation and registration method has been presented for DTs of human-robot safe collaboration [14]. However, these methods demand substantial computational resources and suffer from point cloud offsets caused by agent/object motion [29]. As computer vision algorithms continue to mature, accurate object-level geometric parameters (e.g., pose, shape, and size) become more accessible. Most studies directly update preconstructed 3D models using these parameters. Current research focuses on efficient virtual-real synchronization strategies. A multi-view-driven model update mechanism has been established to mitigate the impact of cluttered backgrounds [6]. A visuo-tactile robotic DT driven by VR-collected human demonstrations has been constructed [7]. A collision detection-based model updating scheme has been proposed for DTs of human-robot collaboration [8]. To reduce data transmission, a task-driven information synchronization method has been presented [9]. However, these methods primarily rely on semantic matching between

detected objects and predefined geometric models, making them vulnerable in scenarios with similar objects or complex backgrounds. In contrast, our approach avoids explicit point cloud reconstruction and semantic matching via a voxel-based retrieval-and-alignment pipeline that learns 3D structure directly from monocular RGB, providing a more efficient and robust alternative for dynamic virtual space updating.

C. Model Retrieval and Alignment for Digital Twin Updating

Model retrieval and alignment, dating to the 1960s [30], combine a database of known 3D models with single RGB images to infer both visible and occluded surfaces of objects. Advances in computer vision have greatly expanded their applications in robot vision, augmented reality, and autonomous driving [18][21]. These methods are typically classified into approximate geometric representation-based methods and CAD model-based methods. The former uses point clouds, meshes and implicit functions as geometric representations of objects, offering compatibility with neural networks and arbitrary resolution, respectively [31][32]. The latter leverages the precision and efficiency of CAD models and provide an effective alternative for scene understanding. IM2CAD uses a synthesis-analysis approach for model retrieval and alignment but suffers from slow inference speeds and assumes that objects are positioned on the ground [17]. To enhance model alignment accuracy, building on this framework, Total3D proposes a comprehensive scene understanding network incorporating layout estimation, object detection, and mesh generation [18]. Mask2CAD optimizes the framework and establishes a two-stage retrieval-alignment method to improve inference speed through task decoupling, but its pose estimation accuracy is limited [19]. Building on this, Patch2CAD refines model retrieval through an image-splitting voting mechanism [20]. However, these methods perform model retrieval and pose estimation separately and rely solely on single RGB image, which leads to scale ambiguity. To address this, ROCA proposes an end-to-end joint reasoning approach that learns 3D canonical coordinate representations from 2D images, facilitating robust joint embedding of 2D and 3D information [21]. FastCAD [40] further adopts contrastive learning and embedding distillation to enable efficient and accurate multi-object CAD retrieval and 9-DoF alignment. To ensure model stability, DiffCAD explores a weakly supervised CAD model retrieval method using disentangled diffusion models and multi-hypothesis sampling [23]. However, diffusion model sampling is time-consuming and can be impractical for DT systems with strict real-time requirements. Subsequently, InCAD [24] significantly improves efficiency over DiffCAD by replacing multi-hypothesis diffusion sampling with deterministic, geometry-aware matching and NOC-based optimization. Additionally, to improve single-view retrieval robustness to unseen shapes, Gen3DRet [33] adopts a two-stage training pipeline—synthetic pretraining on occlusion-rich scenes followed by finetuning on real images. Building on these advances, we integrate a complete 3D voxel representation and a differentiable Top-k retriever into the model

retrieval-and-alignment pipeline, together with a learnable weighted two-stage rotation estimation strategy, to enhance system robustness in complex embodied-agent scenarios involving occlusions and previously unseen objects.

III. OVERVIEW OF *VIREQPHY*

Consider an agent performing embodied tasks by perceiving real-time environments through visual sensors. From scene RGB images, *VirEqPhy* outputs CAD models and corresponding object poses to update the digital space (**DS**). The workflow is illustrated in Fig. 1.

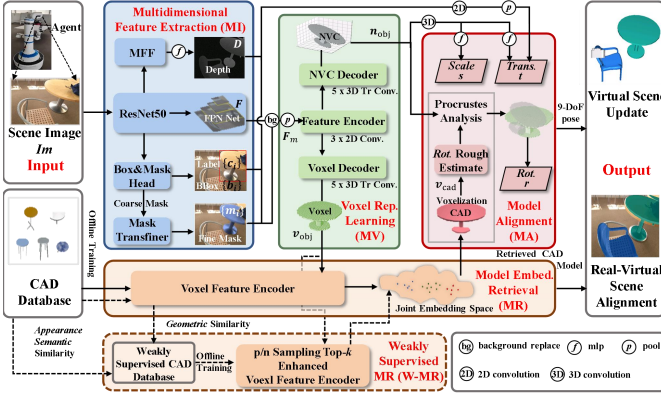


Fig. 1. Overview of *VirEqPhy*. The dashed arrow represents the optional weakly supervised model retrieval process.

First, the multidimensional feature extraction module (**MF**) extracts image representations $\mathbf{F}_I$ (object class, bounding box, mask), focusing on relevant object features while reducing background noise. These features enable subsequent model retrieval and alignment. **Second**, the voxel representation learning module (**VL**) processes these features through a feature encoder to generate semantic embeddings. These are then fed into the voxel decoder and voxel coordinate decoder to predict the standard voxel $\mathbf{v}_{obj}$ and voxel coordinate $\mathbf{n}_{obj}$, respectively, forming the complete voxel representation $\mathbf{F}_V$ of the object. This conversion captures both visible and occluded object parts via 3D voxel grids. **Third**, the model embedding retrieval module (**MR**) converts object voxels into voxel embedding through a voxel feature encoder and maps it into a joint CAD-model space. Using nearest neighbor search (NNS), it retrieves the most similar CAD model cad_{obj} from the database. By embedding both real-world objects and CAD models into a joint feature space, the most similar model from the database can be retrieved without three-dimensional reconstruction. **Lastly**, the model cross-domain alignment module (**MA**) coarsely estimates object’s rotation using the database priors, then refines it via learnable weighted Procrustes analysis between the retrieved CAD model and object voxel coordinates to determine precise rotation $\mathbf{r}$. Position $\mathbf{t}$ and scale s are predicted separately via multidimensional feature regression, yielding the 9-DoF pose $(\mathbf{s}, \mathbf{t}, \mathbf{r})$ of the object. Accordingly, the virtual scene can be updated by incorporating the retrieved CAD model. Cross-domain alignment refines the spatial properties of the retrieved CAD models to match current physical layout, ensuring accurate virtual space updates.

Alternatively, a weakly supervised model retrieval module (**W-MR**) is introduced in a plug-and-play mode to generalize the system to unseen objects. Similar to the supervised setting, **W-MR** matches objects with CAD models by mapping them into a joint embedding space. Differently, it leverages multiple similarities (appearance, semantic, and geometric) as fuzzy labels for matching object-model and optimizes the joint embedding space via a differentiable Top- k algorithm with positive/negative samples. By expanding the model retrieval process, it is capable of retrieving the most relevant CAD models for objects that have not been seen during training.

We develop a comprehensive formula to effectively describe *VirEqPhy*. Let $\mathbf{E} = \{obj_1, ..., obj_{|E|}\}$ and $\mathbf{DB} = \{cad_1, ..., cad_{|\mathbf{DB}|}\}$ denote the set of scene objects and CAD database, respectively, where $|E|$ and $|\mathbf{DB}|$ refer to the number of objects and models. For any object $obj \in \mathbf{E}$ in current scene image $\mathbf{Im}$, its virtual counterpart’s geometric update follows Eq. (1), with $\mathbf{F}_I = \mathbf{MI}(\mathbf{Im})$ and $\mathbf{F}_V = \mathbf{MV}(\mathbf{F}_I)$. Notably, **MR** supports both supervised and unsupervised modes, enabling seamless adaptation to general and unknown-object scenes.

$$\mathbf{DS}(obj) = \mathbf{VirEqPhy}(Im, DB): \begin{cases} cad_{obj} = \mathbf{MR}(F_V, DB) \\ (s, t, r) = \mathbf{MA}(F_I, F_V, cad_{obj}) \end{cases} \quad (1)$$

IV. MULTI-TASK CASCADE LEARNING FRAMEWORK

A. Multidimensional Feature Extraction

First, a multi-scale feature pyramid $\mathbf{F}$ of the scene RGB image is generated using ResNet-50-FPN. Following the methodology of Mask R-CNN [34], bounding boxes $\{\mathbf{b}_i\}$, classes $\{c_i\}$ for each object $\{obj_i\}$ are obtained, where i denotes the index of objects within the agent's field of view. Subsequently, a set of coarse instance masks is generated via semantic segmentation on the bounding box regions using a fully convolutional network. To mitigate background inclusion and detail loss, a Transformer-based Mask Transformer is employed to segment incoherent regions, outputting fine instance masks $\{\mathbf{m}_i\}$. Finally, $\mathbf{F}$ undergoes multi-scale feature fusion to generate a depth feature, processed by a fully connected (fc) network to produce depth map $\mathbf{D}$. For clarity, the subscript i is omitted in the subsequent symbols.

B. Voxel Representation Learning

Effective object feature representation is critical for matching target object with CAD model in the database. Leveraging voxel grids' suitability for 3D information representation, we introduce an encoder-dual decoder framework to estimate voxel representations and voxel coordinates (VC) of objects, mitigating occlusion effects on feature expression. Specifically, the foreground and background of F are separated using the mask $\mathbf{m}$, and the masked feature $F_m \in \mathbb{R}^{256 \times 256}$ is extracted via a pooling layer. The voxel encoder processes these features through a 2D convolutional layer (32 kernels) to generate intermediate features. Building on Pix2Vox++ [35], two voxel decoders are employed to convert intermediate features into a 3D voxel

representation $\mathbf{v}_{obj}$ and VC prediction $\mathbf{n}_{obj}$ with dimensions of $32^3 \times 3$. The voxel decoder comprises five 3D transposed convolutional layers: kernel size 4^2 for the first four layers and 1 for the final layer. The *Sigmoid* function predicts the occupancy probability for each voxel grid. In the VC decoder, the first four convolutional layers are the same as those of the voxel decoder, while the final layer predicts voxel coordinates across three channels.

C. Model Embedding Retrieval

Effective model retrieval hinges on a feature encoder that projects objects and CAD models into a shared embedding space, where objects and its corresponding CAD models (i.e., object-cad associations) exhibit minimal distance, as shown in Fig. 2 (a). Unlike prior works [17][18][19][20], we construct a joint embedding space using complete voxel representations of objects and CAD models. First, CAD models are voxelized to $\mathbf{v}_{cad}$ (similar to $\mathbf{v}_{obj}$), bridging feature distribution gaps. Next, a twin network with four 3D convolutional blocks serves as the voxel feature encoder, processing both $\mathbf{v}_{obj}$ and $\mathbf{v}_{cad}$ to generate embedding vectors $\mathbf{f}_{obj}^{vox} \in \mathbb{R}^{256}$ and $\mathbf{f}_{cad}^{vox} \in \mathbb{R}^{256}$ mapped to the joint space. Each 3D convolution block includes a downsampling block and a residual block, as depicted in Fig. 2 (b). Finally, the cosine distance is used to retrieve the closest $\mathbf{f}_{cad}^{vox}$ to $\mathbf{f}_{obj}^{vox}$ via NNS, with the corresponding CAD model taken as the prediction. This process is trained using ground-truth object-cad associations from the database. To enhance retrieval accuracy, three techniques are employed: 1) A hard sample mining method [18] is applied during training to capture challenging-to-distinguish features; 2) Only CAD models matching the object's class are considered during retrieval; 3) Following Patch2CAD [20], 3D Mask occlusion is applied to 32^3 -dimensional voxel spaces to improve matching of incomplete features.

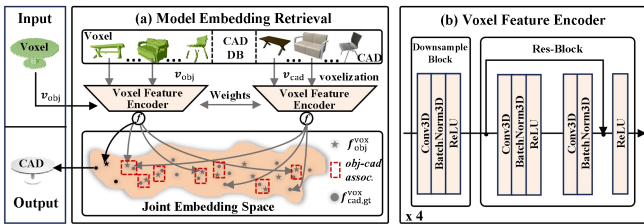


Fig. 2. Model retrieval network. The gray line represents the object-cad associations samples mapping process.

D. Model Cross-Domain Alignment

Accurate estimation of scale $\mathbf{s} \in \mathbb{R}^3$, position $\mathbf{t} \in \mathbb{R}^3$, and rotation $\mathbf{r} \in \mathbb{R}^4$ of objects in RGB images is fundamental for model cross-domain alignment. These denote the scaling factors along each axis, the coordinates of the 3D object center, and the rotation matrix transforming retrieved CAD model's standard orientation to current object orientation, respectively.

1) Scale and Position Estimation Since VC encode both pose and scale information, a 3D CNN downsampling layer is employed in scale estimation to process $\mathbf{n}_{obj}$, producing intermediate feature $\mathbf{n}'_{obj}$ with dimensions of $16^3 \times 9$. After

unfolding this feature, a three-layer fc network is used to predict the scale values specific to each objects class, with final scale $\mathbf{s}$ determined by object class. For position estimation, an initial position $\mathbf{t}_0$ is derived by processing the masked depth map $\mathbf{D}_m$ through a 2D convolutional layer followed by average pooling. Similar to scale estimation, an fc network—incorporating object class information—is used to estimate the positioning offset $\Delta \mathbf{t}$, yielding the final position prediction $\mathbf{t} = \mathbf{t}_0 + \Delta \mathbf{t}$. Notably, indoor scenarios (e.g., offices, homes) exhibit consistent depth distributions and object layouts. Consequently, depth estimation can mitigate scale ambiguity inherent in RGB image-based methods.

2) Rotation Estimation Unlike scale and position, rotation estimation is theoretically unaffected by scale ambiguity but highly sensitive to occlusion. Observing that object postures stabilize around fixed orientations due to physical logic constraints in indoor scenarios, this section proposes a two-stage rotation estimation method. First, the *K-medoid* algorithm clusters 8 most common pose centers per object category. VC of the object is then classified into these clusters via cross entropy loss, yielding the coarse rotation prediction $\mathbf{r}_0$. Second, the retrieved CAD model is voxelized to voxel coordinates $\mathbf{n}_{cad}$, which is rotated by $\mathbf{r}_0$ to produce $\mathbf{n}'_{cad}$, minimally differing from $\mathbf{n}_{obj}$. Finally, using weighted Procrustes analysis, the orthogonal rotation $\mathbf{r}'$ between $\mathbf{n}'_{cad}$ and $\mathbf{n}_{obj}$ is computed, as shown in Eq. (2), where $\mathbf{L}$ contains occupancy grids present in both $\mathbf{n}'_{cad}$ and $\mathbf{n}_{obj}$, $\tilde{w}(l)$ denotes the predicted probability for the l -th occupancy grid in $\mathbf{L}$, and $\odot$ indicates the element-wise product. The final rotation prediction is $\mathbf{r} = \mathbf{r}' + \mathbf{r}_0$.

$$\mathbf{r}' = \underset{\mathbf{r}_i}{\operatorname{argmin}} \sum_{l=1}^{|\mathbf{L}|} \tilde{w}(l) \left(\mathbf{n}_{obj}(l) - \mathbf{r}' (\mathbf{n}'_{cad}(l) \odot \mathbf{s}) \right)^2 \quad (2)$$

E. Multi-Task Learning Joint Loss

In **MI**, the loss mainly consists $\mathcal{L}_{2D}$ (class, bounding box, coarse/fine mask estimation losses) and $\mathcal{L}_{depth}$ (*BerHu*-based depth estimation loss). In **MV**, the probability p of each voxel's occupancy grid in $\mathbf{v}_{obj}$ and the binary CAD voxel representation $\mathbf{v}_{cad}$ serve as outputs and supervision, respectively. The VC supervision $\mathbf{n}_{obj,gt}$ is derived by scaling and rotating voxelized CAD coordinates using ground-truth scale $\mathbf{s}_{gt}$ and rotation $\mathbf{r}_{gt}$. The loss is expressed by Eq. (3).

$$\mathcal{L}_{MV} = \frac{1}{n} \sum [(\mathbf{v}_{cad} \log(p) + (1 - \mathbf{v}_{cad}) \log(1 - p)) + \|\mathbf{n}_{obj} - \mathbf{n}_{obj,gt}\|] \quad (3)$$

In **MR**, the voxel feature encoder is trained using positive (matched) and negative (unmatched) samples of voxelized CAD models. The noise contrastive estimation loss is chosen to evaluate the effectiveness of the joint embedding space, which is expressed by Eq. (4).

$$\mathcal{L}_{MR} = - \sum \log \left[\frac{\operatorname{dis}(\mathbf{f}_{obj}^{vox}, \mathbf{f}_{cad,gt}^{vox,+})}{\operatorname{dis}(\mathbf{f}_{obj}^{vox}, \mathbf{f}_{cad,gt}^{vox,+}) + C \sum \operatorname{dis}(\mathbf{f}_{obj}^{vox}, \mathbf{f}_{cad,gt}^{vox,-})} \right] \quad (4)$$

where $f_{cad,gt}^{vox,+}$ and $f_{cad,gt}^{vox,-}$ represent the embeddings of positive and negative CAD samples, respectively, C is a constant used to adjust the negative samples influence, and dis denotes the cosine distance operation.

In **MA**, the loss $\mathcal{L}_{MA}$ primarily comprises the scale, position, and rotation estimation loss, each weighted by 1, 0.5 and 1.5, respectively. The total loss function $\mathcal{L}_{total}$ combines multi-task objectives with dynamically weights adjustment, is defined as Eq. (5), where λ represents the loss weights adjusted via dynamic weight averaging (DWA) method [36].

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{2D} + \lambda_2 \mathcal{L}_{depth} + \lambda_3 \mathcal{L}_{MV} + \lambda_4 \mathcal{L}_{MR} + \lambda_5 \mathcal{L}_{MA} \quad (5)$$

F. Supervised CAD Database Construction

Considering the object occlusion and cluttered backgrounds exists in ScanNet dataset, we follow the Scan2CAD approach [37], aligning the image dataset ScanNet with the CAD model dataset ShapeNet to create a supervised Scan2CAD-based dataset. This dataset includes 14,225 objects from 1,506 video sequences (706 indoor scenarios) in ScanNet, matched with 3,049 CAD models from ShapeNet. It covers 9 common object types: *bed, sofa, chair, bin, cabinet, display, table, bookshelf, and bathtub*. Annotated labels include object-cad associations, 9-DoF poses of the CAD models, object classes, masks, and depth images.

V. WEAKLY SUPERVISED LEARNING-ENHANCED MODEL RETRIEVAL

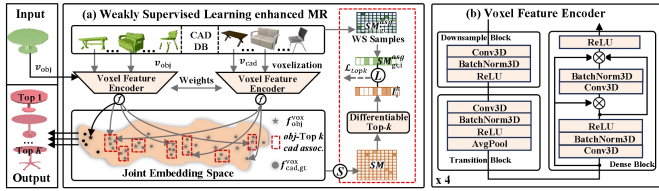


Fig. 3. Weakly supervised learning-enhanced MR network. The gray line represents the samples mapping process of object-to-Top-k cad associations and $\otimes$ means concatenation operation.

The previous section introduced an end-to-end framework for virtual-physical equivalence that leverages voxels as a medium to achieve CAD model retrieval based on 2D images. It depends critically on large-scale, high-quality manually-annotated datasets. However, the high cost and subjectivity of manual labeling yield limited effective samples, typically covering only partial object instances. This restricts the proposed framework's generalization capability in complex scenarios with unseen objects. To overcome this, we introduce a weakly supervised retrieval module integrated into the framework, as shown in Fig. 3. A fuzzy mapping mechanism is designed for object-cad associations using multiple similarities as weak labels, forming a large-scale weakly supervised dataset. Additionally, a novel differentiable Top-k algorithm—incorporating both positive and negative samples—is employed to guide the construction of a joint embedding space, retrieving the K most similar CAD models for a given object. Without exhaustive and manual labels, this approach supports deployment across diverse supervision

levels, enabling the digital twin system to scale efficiently in real-world applications.

A. Weakly Supervised CAD Database Construction Guided by Multiple Similarities

Comprehensive object-CAD similarity measures are essential for constructing reasonable weakly supervised samples. In this section, the supervised dataset from the previous section is expanded to 20 classes (over 20,000 instances) by incorporating appearance, semantic, and geometric similarities (Fig. 4). Although the dataset includes atypical objects like airplanes and cars, experiment results demonstrate that adding these objects can enhance the model's generalization capability.

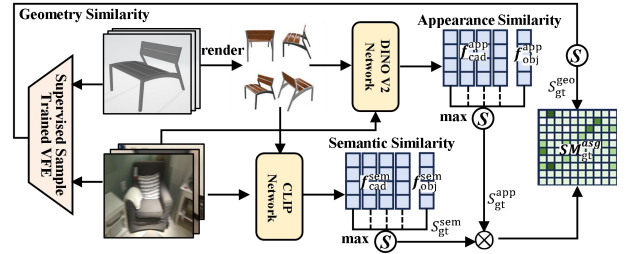


Fig. 4. Construction of weakly supervised dataset based on multiple similarities.

1) Appearance Similarity (S_{gt}^{app}) Appearance similarity quantifies the similarity between objects and CAD models from a 2D visual perspective. With the emergence of visual large models such as SAM [38] and DINOv2 [39], universal visual features applicable across tasks without fine-tuning are easily available. In this section, each CAD model is offline-rendered from R viewpoints to generate a set of 2D images, from which pre-trained model DINOv2 extracts visual information as appearance feature of the CAD model, denoted as $f_{cad}^{app} \in \mathbb{R}^{256 \times R}$. Similarly, the appearance feature $f_{obj}^{app} \in \mathbb{R}^{256}$ of the scene object is obtained. The cosine similarity, which is robust to data scale and computationally efficient, is then employed to calculate the appearance similarity, defined as $S = (f_{obj}/\|f_{obj}\|)(f_{cad}/\|f_{cad}\|)^T \in (-1, 1)$. Maximum similarity across viewpoints is the final appearance similarity.

2) Semantic Similarity (S_{gt}^{sem}) Semantic similarity assesses textual relationship between objects and CAD models. In this section, a pre-trained image-language large models, CLIP, is used to extract semantic features of object images and rendered CAD images, denoted as $f_{cad}^{sem} \in \mathbb{R}^{512}$ and $f_{obj}^{sem} \in \mathbb{R}^{512}$, respectively. To leverage the complementary appearance-semantic information, the semantic features are concatenated with appearance features to calculate a combined cosine similarity S_{gt}^{as} .

3) Geometric Similarity (S_{gt}^{geo}) Geometric similarity quantifies the structural and morphological similarity between objects and CAD models. Unlike IoU-based methods, this section uses voxel feature distances to more precisely evaluate 3D geometric differences between the object and the CAD model. Notably, the **MV** and **MR** modules, learned from the supervised data, are utilized to calculate the voxel embeddings f_{obj}^{vox} and f_{cad}^{vox} of the object and CAD model, respectively.

The corresponding geometric similarity S_{gt}^{geo} is then calculated. Finally, the comprehensive similarity $S_{gt}^{asg} = \alpha_{as} S_{gt}^{as} + \alpha_g S_{gt}^{geo}$ serves as the object-cad mapping label, forming a normalized similarity matrix $\mathbf{SM}_{gt}^{asg}$ for all objects and CAD models. Each row of $\mathbf{SM}_{gt}^{asg}$ represents an *object-to-Top-k cad association*.

B. Top-k Enhanced Voelx Feature Encoder

To learn *object-to-Top-k cad associations* mapping, this section designs a 3D DenseNet-based voxel feature encoder for similarity feature extraction, building on the voxel feature encoder introduced in Section III, as shown in Fig. 3(b). However, the weakly supervised dataset contains only similarity values between objects and various CAD models, making conventional loss functions unsuitable for optimization. The Top- k algorithm search for the top K elements from a set and provides a means for constructing a weakly supervised loss function. Let $\mathbf{SM}$ be the normalized similarity matrix between objects and CAD models predicted by the voxel feature encoder, where $\mathbf{SM}_{i,j}$ denotes the similarity between the i -th object and the j -th CAD model. For $\mathbf{SM}_i$, we derive its indicator vector $\mathbf{I}_i$ using an augmented Top- k algorithm, which consists of binary values (0 and 1) and is defined by Eq. (6). Here, a value of 1 indicates that the similarity ranks among top K similarities of the row.

$$\mathbf{I}_{i,j} = \begin{cases} 1, & \mathbf{SM}_{i,j} \text{ in top } K \text{ of } \mathbf{SM}_i \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Considering the non-differentiability of the above operation, we model the Top- k -based retrieval problem as a linear programming problem using optimal transport theory, as shown in Eq. (7) [40]. The objective is to select the top K CAD models most similar to the object, where $|\mathbf{DB}|$ denotes the number of CAD models, $\langle \cdot, \cdot \rangle$ represents the inner product operation, and $\mathbf{1}$ is an all-one vector. The model retrieval process can be formulated end-to-end as shown in Eq. (8), where $\mathbf{DB}^K$ is the set of the retrieved K CAD models, and *Score* and *Similarity* denote the similarity scores and similarity values, respectively. These similarity scores can be directly used to compute gradients and update the encoder parameters.

$$\max \langle \mathbf{I}_i, \mathbf{SM}_i \rangle, \text{ s.t. } \mathbf{I}_i \in \mathbb{R}^{|\mathbf{DB}|}, \mathbf{I}_{i,j} \geq 0, \mathbf{I}_i \mathbf{1} = K \quad (7)$$

$$\arg\max_{\mathbf{DB}^K} \text{Score}(\text{obj}_i, \mathbf{DB}^K) = \frac{1}{K} \sum_{k=1}^K \text{Similarity}(\text{obj}_i, \mathbf{DB}_k^K) \quad (8)$$

To ensure that the learned similarity scores align with the ground-truth similarity scores, the difference between predicted $\mathbf{SM}$ and $\mathbf{SM}_{gt}^{asg}$ should be minimized. Let $\mathbf{I}_i^k$ be the indicator vector computed using the differentiable Top- k algorithm, then the loss function can be defined as Eq. (9).

$$\mathcal{L}_{topk}^+ = -\frac{1}{K} \sum_{k=1}^K (\mathbf{I}_i^k \cdot \mathbf{SM}_{gt,i}^{asg}) \quad (9)$$

However, this process only considers positive samples. To attract similar samples while repelling dissimilar ones—preventing over-concentrated embeddings, we utilize

$\mathbf{1} - \mathbf{SM}_{gt,i}^m$ to identify the $2K$ least similar negative samples, from which $\lambda_{np}K$ negative samples are selected for joint training with positive samples, where $\lambda_{np} = 0, 1, 2$ and $m \in \{\text{as}, \text{geo}\}$. Note that: 1) Similarity normalization enables this approach; and 2) Maximizing differences from appearance-semantic fusion or geometric perspectives ensures valid $2K$ negative sampling. The loss function for negative samples is defined as Eq. (10), yielding the total loss function is $\mathcal{L}_{topk} = \mathcal{L}_{topk}^+ + \mathcal{L}_{topk}^-$.

$$\mathcal{L}_{topk}^- = \frac{1}{2K} \sum_{m \in \{\text{as}, \text{g}\}} \sum_{k=1}^K (\mathbf{I}_i^k \cdot (\mathbf{1} - \mathbf{SM}_{gt,i}^m)) \quad (10)$$

VI. EXPERIMENT AND ANALYSIS

To validate the proposed method's effectiveness, we conduct three experiments: 1) Evaluating the virtual scene updating capability of *VirEqPhy* on the supervised dataset; 2) Testing the model retrieval performance of **W-MR** on the weakly supervised dataset, particularly for unseen objects; and 3) Exploring the feasibility of integrating *VirEqPhy* into real-world embodied agent systems. **More experiments are provided in Appendix.**

A. Performance of Virtual Scene Updating

1) Experimental Settings Using the supervised dataset constructed in Section IV. F, *VirEqPhy* is trained with Pytorch3D on a 32 GB Tesla V100 GPU. The optimizer momentum size, initial learning rate, maximum iteration count, and batch size are set to 0.9, 0.001, 90000, and 32, respectively. In multi-task learning, the initial loss weights ($\lambda_1 = 0.5$, $\lambda_2 = 0.5$, $\lambda_3 = 1$, $\lambda_4 = 1$, $\lambda_5 = 2$) are dynamically adjusted via DWA after 10000 iterations. From 1,506 video sequences in the supervised dataset, over 24,000 RGB images are downsampled with a factor 100 (default), and then split into training and testing sets in an 8:2 ratio.

2) Evaluation Metrics. We adopt the Scan2CAD criterion [37], which is widely used in CAD retrieval and pose estimation tasks. **Retrieval Accuracy:** Considering the non-one-to-one correspondence between objects and CAD models in ScanNet, we use average precision (AP) to assess the model retrieval accuracy, following Mesh R-CNN approach [32]. It is computed as the harmonic mean of precision and recall over the triangular mesh sampling areas of the predicted CAD model and its ground truth under a chamfer distance threshold. **Alignment Accuracy:** For a correctly retrieved CAD model, the model is considered correctly aligned with the object if the errors in position, rotation angles in all directions, and scale must not exceed 20 cm, 20°, and 20%, respectively. **Retrieval-Aware Alignment Accuracy:** Beyond standard alignment accuracy, this metric also requires the retrieved CAD model to match the ground truth, so only one-to-one object-CAD pairs in ScanNet are evaluated.

3) Baseline Methods We select several state-of-the-art methods or representative methods as baselines. **Mask2CAD** (ECCV'20) [19]: A widely adopted two-stage baseline with a 2D images-based joint embedding space. **Patch2CAD** (ICCV'21) [20]: A patch-wise retrieval approach that

strengthens local image-shape matching. **Total3D** (CVPR’20) [18]: A canonical synthesis-analysis pipeline for full-scene inference under layout assumptions. **ROCA** (CVPR’22) [21]: An end-to-end method using a 3D-coordinate joint embedding with Procrustes-based alignment. **FastCAD** (ECCV’24) [22]: An end-to-end approach leveraging temporal fusion for scans and videos via contrastive learning and distillation. **DiffCAD** (TOG’24) [23]: A weakly supervised, diffusion-based probabilistic framework. **InCAD** (ICCV’25) [24]: A zero-shot, weakly supervised alignment method built on ROCA’s retrieval backbone with geometry-aware matching. Furthermore, to verify the effectiveness of various components in proposed method and the impact of input modality and training data capacity on system performance, we conduct ablation experiments. **Ours-base**: A basic multi-task architecture with single-stage rotation estimation, 2D feature encoding, chamfer distance, fixed task weights and coarse masks. **Ours**: The complete multi-task architecture featuring two-stage rotation estimation (**R**), voxel representation learning (**V**), joint embedding space (**J**), dynamic task weights (**M**) and refined masks (**T**), with depth estimation. The downsampling factor of the video sequence is 100. **Ours-rgb**: Replaces the depth estimation with real depth from an RGB-D camera. **Ours-50**: Reduces the downsampling factor from 100 to 50, thereby increasing the number of training samples.

To ensure a fair comparison, we 1) restrict **FastCAD** to a single-image setting under the video mode, 2) report a single-hypothesis result for **DiffCAD**, selected by the ground-truth pose, from its multi-hypothesis outputs, 3) report retrieval accuracy only for **Total3D** and **Patch2CAD** and provide ground-truth depth to the network, since they both support shape retrieval only; 4) report alignment accuracy only for **InCAD**, since it uses **ROCA** for retrieval.

4) **Results** Table I and I report the Retrieval Accuracy and Alignment Accuracy of the various methods, respectively. From Table I, overall, our method achieves the best high-threshold retrieval and the second-best mean retrieval, reflecting that the voxel-based joint embedding space favors tighter shape-pose matches at stricter IoU thresholds. **DiffCAD** is highly competitive under weak supervision and tends to lead on categories with regular, stable geometry (e.g., box-like furniture), where probabilistic multi-hypothesis matching readily captures near-duplicate shapes. **Patch2CAD**, with its patchwise feature embedding, attains solid retrieval accuracy, particularly on smaller or edge-dominated objects such as *bin* and *display*. **ROCA** benefits from a 3D coordinate joint embedding that improves robustness over classic two-stage methods (e.g., **Mask2CAD**), but without explicit pose refinement in the retrieval objective, its gains remain limited. As shown in Table I, our method achieves the best class-level and second-best instance-level alignment, indicating stronger global spatial consistency under the joint position/rotation/scale criterion. It is particularly effective on geometry-driven categories (*cabinet*, *display*, *table*, *book-shelf*), owing to voxelized geometry and a two-stage rotation scheme that reduces pose ambiguity and couples

retrieval with pose for tighter alignment. Notably, **InCAD** performs strongly on shape-rich categories (e.g., *sofa*, *chair*) via geometry-aware learning but underperforms on box-like categories with uniform shape (e.g., *cabinet*). **DiffCAD** remains broadly stable across categories — especially *bed* and *book-shelf* — owing to its diffusion-based probabilistic modeling, which improves global alignment consistency.

Table I Retrieval Accuracy (%) of various methods. **AP50**, **AP75**, and **APm** represent the average retrieval accuracy at chamfer distance thresholds of 50%, 75%, and the average over the range from 50%-95%, respectively. The blue background indicates the top three values. # and * represent the results of the original report and the reproduced results respectively.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	APm (mean)	AP50 (mean)	AP75 (mean)
Total3D#	1.9	4.3	1.5	0.8	0.1	0.0	0.7	2.1	1.4	6.3	0.2
Mask2CAD#	14.2	13.0	13.2	7.5	7.8	5.9	2.9	3.1	8.4	22.5	4.9
Patch2CAD#	18.8	12.4	17.6	7.5	8.6	10.8	3.3	3.3	10.3	20.6	6.6
ROCA*	15.6	11.7	17.1	6.5	8.8	9.2	5.4	3.5	10.0	19.4	5.5
FastCAD*	15.5	12.0	17.5	6.4	8.2	8.5	4.8	3.7	9.8	20.5	5.5
DiffCAD*	20.2	11.6	20.1	6.0	10.3	10.7	5.5	5.3	11.6	23.9	6.6
Ours	19.1	13.0	21.9	7.2	9.4	10.5	5.6	3.8	11.3	24.2	6.7

Table II Alignment Accuracy (%) of various methods. Class and Instance represent the average accuracy over the class and instance, respectively. The blue background indicates the top three values. # and * represent the results of the original report and the reproduced results respectively.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	bathub	class	instance
Mask2CAD*	2.9	5.3	30.9	25.9	5.4	17.3	7.1	3.8	8.3	11.9	17.9
ROCA#	10.0	15.9	41.0	29.3	15.8	30.4	14.6	14.2	22.5	21.5	27.4
FastCAD*	17.7	19.1	44.9	30.6	16.1	24.7	17.8	16.5	20.4	23.2	27.6
DiffCAD*	19.5	23.3	45.7	28.9	18.4	25.2	23.2	19.5	20.7	23.8	28.8
InCAD#	18.6	38.1	49.3	22.8	9.2	24.1	16.5	12.7	16.7	23.1	30.1
Ours	16.3	29.8	48.7	27.8	18.6	31.2	24.6	19.5	23.3	24.4	29.6

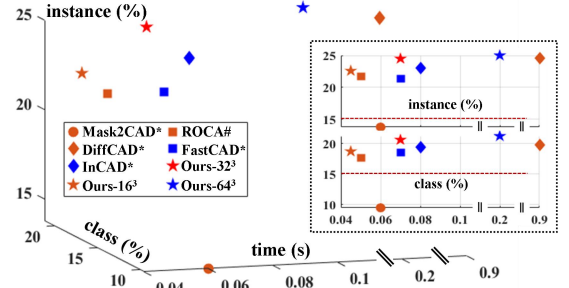


Fig. 5. Retrieval-Aware Alignment Accuracy (%) and inference time (per image) of various methods. **Ours-16³**, **Ours-32³** and **Ours-64³** denote voxel sizes of 16³, 32³ and 64³, respectively. Methods with identical markers denote the same methodological family.

Fig. 5 reports the Retrieval-Aware Alignment Accuracy and mean inference time per image across methods. Our method traces a clear accuracy-latency frontier: **Ours-32³** (default) delivers strong class/instance accuracy at ~0.07 s/img, while **Ours-64³** (larger voxel grid) maximizes accuracy at ~0.20 s/img, and **Ours-16³** (smaller grid) maximizes speed (~0.05 s/img) with a modest drop. Moreover, among End-to-end methods, **ROCA** attains the lowest latency, while **FastCAD** adds contrastive learning and embedding distillation, improving accuracy while retaining real-time speed. For weakly supervised methods, **DiffCAD** is competitive in instance accuracy yet slower (~0.9 s/img) due to diffusion sampling, whereas **InCAD** remains accurate and low-latency, benefiting from **ROCA**’s retrieval module. Overall, our method **VirEqPhy** excels in model retrieval and alignment, and shows a clean accuracy-latency trade-off. Note that

Total3D (synthesis-analysis) and **Patch2CAD** (two-stage) lack a complete retrieval-and-alignment pipeline; for fairness, we exclude them from this comparison. For our default configuration (**Ours-32^s**), the ~ 0.07 s/img runtime consists of ~ 15 ms for feature extraction, 30 ms for voxel learning, 10 ms for model retrieval, and 15 ms for alignment, so the overall latency is dominated by voxel encoding while retrieval and alignment are lightweight.

The ablation results in Table III further indicate that: 1) Two-stage rotation estimation significantly improves model alignment accuracy compared with using only fine rotation estimation (**Ours-base** vs **Ours-R**). This gain is attributed to reduced rotational ambiguity during the coarse stage (e.g., chairs typically facing tables), and this module has the most pronounced impact on inference performance; 2) 3D voxel representation enhances model retrieval compared with 2D features because it better discriminates objects with subtle geometric differences, such as *chair*, *sofa*, *cabinet* (**Ours-R** vs **Ours-R/V**); 3) a learnable joint embedding space further improves model retrieval performance, especially for objects with regular geometric structures (e.g., *bed* and *cabinet*), which may be attributed to its more intuitive geometric similarity measurement (**Ours-R/V** vs **Ours-R/V/J**); and 4) dynamic task weights effectively balance the different tasks, while refined masks further optimize performance (**Ours-R/V/J** vs **Ours**). Extended experiments reveal that increasing the training data capacity can improve model performance, but the gain is limited (**Ours-rgb** vs **Ours**), because too many consecutive frames introduce redundant or weakly informative training data. In addition, the improvement brought by real depth is also limited (**Ours-50** vs **Ours**), possibly due to sensor noise. These results validate the stability and effectiveness of the proposed **VirEqPhy** design.

Table III Retrieval-Aware Alignment Accuracy (%) for various ablations. The underline value indicates the optimal result.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	bathtub	class	instance
Ablations – System components											
Ours-base	5.4	2.6	22.9	22.6	4.6	12.1	8.4	4.6	7.5	10.1	14.5
R	10.4	9.1	26.5	<u>26.0</u>	11.5	19.2	15.2	12.8	16.7	16.4	20.2
R V	14.1	14.0	32.4	23.5	15.3	20.4	16.0	13.2	18.2	18.6	22.3
R V J	<u>17.2</u>	13.5	30.3	22.9	<u>18.3</u>	21.1	20.4	14.1	18.0	19.6	23.1
R V J M	15.8	12.9	34.1	24.7	17.8	<u>22.0</u>	19.8	14.1	20.2	20.2	23.7
Ours	16.1	<u>13.8</u>	<u>35.6</u>	25.5	17.9	21.4	<u>20.7</u>	<u>14.7</u>	<u>20.9</u>	<u>20.6</u>	<u>24.5</u>
Ablations – Input modality & Training data capacity											
Ours-rgb	<u>17.0</u>	<u>14.3</u>	35.4	<u>25.7</u>	<u>18.5</u>	21.2	<u>20.9</u>	14.5	20.7	<u>21.0</u>	24.8
Ours-50	16.5	13.9	<u>35.8</u>	25.4	18.4	21.3	20.5	14.5	<u>21.2</u>	20.8	<u>25.1</u>
Ours	16.1	13.8	35.6	25.5	17.9	<u>21.4</u>	20.7	<u>14.7</u>	20.9	20.6	24.5

Fig. 6 visualizes scene updates across single- and multi-object scenarios. It can be seen that even in complex scenarios, our method outperforms almost all baselines, preserving object quantity, layout, and pose with fewer scale/boundary errors. Moreover, consistent with quantitative results, **DiffCAD** can also produce visually consistent shapes with minor pose/scale errors. Specifically, in the third and fourth instances (cluttered room with partially occluded *table*, *chair*, and *cabinet*), baselines often yield fragmented or misaligned CAD models, whereas ours matches the number and position of real objects; in the seventh instance (overexposed lighting), baselines often generate incorrect CAD models or fail to align them with the real objects, while

ours maintains relatively accurate updates. Notably, **InCAD**, built on **ROCA**, achieves more accurate alignments than **ROCA**. Both quantitative and qualitative results confirm **VirEqPhy**'s effectiveness in achieving geometric equivalence between virtual and physical spaces.

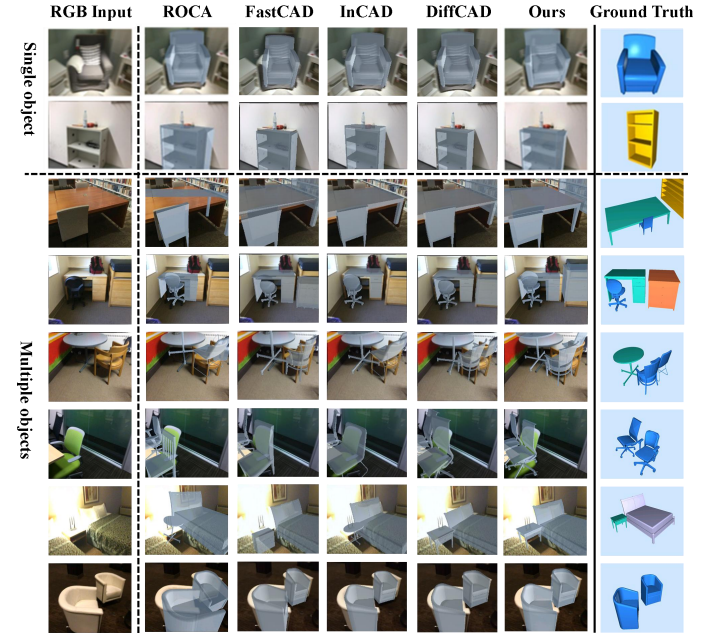


Fig. 6. Qualitative results in scenarios with single and multiple objects.

B. Performance of Model Retrieval of W-MR

1) Experimental Settings The **W-MR** module is trained on the weakly supervised dataset with a batch size of 32. For each image, objects with the highest confidence are selected for training. During each iteration, the top two most similar models for each object are collected to form a model library containing 60 models for retrieval (with $\alpha_{as}=0.4$ and $\alpha_g=0.6$).

2) Evaluation Criteria The evaluation criteria used in this section are as follows. **Retrieval Accuracy**: As defined previously. **Top-1 Retrieval Accuracy (Accu1)**: The top-1 retrieved model matches the object. **Top-5 Retrieval Accuracy (Accu5)**: At least one of the top-5 retrieved models matches the object. **IoU1**: Intersection over union between the top-1 retrieved model and the ground truth. **IoU5**: Extension of IoU1 to the top-5 retrieved models.

3) Baseline Methods Three methods, **Mask2CAD** (ECCV'20) [19], **Patch2CAD** (ICCV'21) [20] and **DiffCAD** (TOG'24) [23], known for their strengths in model retrieval, are chosen as baseline methods. **Ours-supervised**: Supervised learning methods described earlier. A novel two-stage model retrieval training pipeline, **Gen3DRet** (3DV'24) [33], is introduced: Synthetic pretraining followed by real-data fine-tuning for embedding learning. It excels in complex scenarios with unseen objects. Furthermore, we conduct ablation experiments on loss functions, sample ratios, and multiple similarity measures to validate the effectiveness of the proposed module. **Ours-MSE**: Mean squared error loss between the Top-1 retrieved model embeddings and object embeddings. **Ours-CL**: Noise contrastive estimation loss between highest/lowest similarity model embeddings and

object embeddings. **Ours-Top-1** and **Ours-Top-5**: Top- k loss with positive samples only ($K=1, 5$). **Ours-p/n Top-5**: Top- k loss with both positive and negative samples.

4) **Results** Table IV reports Retrieval Accuracy of various methods (*bathtub* excluded for comparability). Even without precise object-CAD supervision, the **W-MR** module achieves superior retrieval performance across most categories, outperforming all baselines. This is due to Top- k positive/negative training under weak supervision, which sharpens fine-grained discrimination; gains are more pronounced for classes with rich intra-class variation (e.g., *chair*, *cabinet*) than for simpler or more rigid items (e.g., *bin*). Furthermore, note that 1) although **DiffCAD** is also a weakly supervised learning method, its diffusion-based matching can introduce sampling variance and synthetic-data bias, thereby constraining its performance in strict-overlap retrieval; and 2) **Gen3DRet** benefits from two-stage training and multi-view shape embeddings, giving it outstanding performance in high-threshold retrieval. However, this method comes with increased training cost.

Table IV Retrieval Accuracy (%) of various methods. The blue background indicates the top three values. # and * represent the results of the original report and the reproduced results respectively.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	Apm (mean)	AP50 (mean)	AP75 (mean)
Mask2CAD#	14.2	13.0	13.2	7.5	7.8	5.9	2.9	3.1	8.4	22.5	4.9
Patch2CAD#	18.8	12.4	17.6	7.5	8.6	10.6	3.3	3.3	10.3	20.6	6.6
DiffCAD*	20.2	11.6	20.1	6.0	10.3	10.7	5.5	5.3	11.6	23.9	6.6
Ours-supervised	19.1	13.0	21.9	7.2	9.4	10.5	5.6	3.8	11.3	24.2	6.7
Gen3DRet* (Top-5)	23.0	12.4	22.3	6.6	12.1	10.9	5.8	5.5	12.2	28.9	7.1
Ours-p/n Top-5	23.2	12.7	22.9	6.3	12.7	10.8	6.4	5.4	12.5	29.2	7.0

Table V Retrieval Accuracy (%) under different combinations of similarity measures. The underline value indicates the optimal result. $\lambda_{np} = 0$, $\lambda_{np} = 1$ and $\lambda_{np} = 2$ respectively represent the ratios of negative samples to positive samples as 0 (all positive samples), 1, and 2.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	Apm (mean)
Ablations - Loss design & Samples ratios									
Ours-MSE	13.7	9.5	17.6	5.8	11.6	7.5	4.9	3.7	9.3
Ours-CL	20.4	11.8	22.5	6.3	13.0	9.6	4.7	4.8	11.7
Ours-Top-1 ($\lambda_{np}=0$)	22.1	12.3	21.7	6.5	12.3	10.2	4.8	5.2	12.0
Ours-Top-5 ($\lambda_{np}=0$)	23.6	12.3	21.9	6.5	11.9	10.5	5.9	5.2	12.2
Ours-p/n Top-5 ($\lambda_{np}=1$)	23.5	12.8	22.7	6.2	12.5	10.8	6.3	5.4	12.4
Ours-p/n Top-5 ($\lambda_{np}=2$)	23.2	12.7	22.9	6.3	12.7	10.8	6.4	5.4	12.5
Ablations - Similarity combination									
Ours-p/n Top-5 w/o Geo.	8.6	5.3	10.1	5.7	9.6	6.5	4.9	3.7	6.8
Ours-p/n Top-5 w/o App.	11.8	7.5	15.0	6.2	13.1	7.9	5.8	4.0	8.9
Ours-p/n Top-5 w/o Sem.	21.5	12.9	21.8	6.5	12.9	9.4	6.0	5.1	12.0
Ours-p/n Top-5	23.2	12.7	22.9	6.3	12.7	10.8	6.4	5.4	12.5

The ablation experiment results are presented in Table V. Compared to traditional loss optimization, the Top- k loss optimization enhances the overall retrieval performance of the **MR** module (**Ours-Top** vs **Ours-MSE/CL**). Incorporating Top- k loss with positive and negative samples yields the best performance, further enhancing fine-grained discrimination, particularly on *sofa* and *display* (**Ours-p/n Top** vs **Ours-Top**). Furthermore, as the number of negative samples increases, retrieval accuracy gradually improves with diminishing returns; the gains are concentrated in high intra-class variation categories, such as *chair* ($\lambda_{np}=0$ vs $\lambda_{np}=1$ vs $\lambda_{np}=2$). However, these improvements are not significant, potentially due to a suboptimal balance between positive and negative examples and the introduction of noise from negative samples,

which complicates the learning process. For similarity design, the model achieves the highest retrieval accuracy when integrating all three similarities. Removing geometric, appearance and semantic similarities leads to accuracy reductions of 5.7, 3.6 and 0.5 (**w/o Geo. vs w/o App. vs w/o Sem.**), respectively. This indicates that the model is most sensitive to geometric information. This is because geometric similarity quantifies the structural and morphological similarity between objects and CAD models, which is essential for object-CAD matching. In particular, our work utilizes 3D voxel features as the geometric similarity measure.

To assess **W-MR**'s generalization on unseen objects, the weakly supervised dataset is restructured: using *bed*, *bin*, *cabinet*, *display* and *bookshelf* as seen objects for training, while *sofa*, *chair* and *table* as unseen objects for evaluation. The experimental results are shown in Fig. 7. On seen objects, our method outperforms all baselines, which is consistent with the previous results; that is, with familiar categories, the Top- k objective and p/n sampling enhance the voxel-anchored joint embedding for object-CAD matching. On unseen objects, our method exhibits stronger Top-5 performance, whereas Top-1 performance is near-optimal, indicating that it better covers "similar but not identical" candidate models. Notably, the performance of **Ours-Supervised** degrades markedly on unseen objects, while **Ours-p/n Top-5** significantly alleviates this drop across all metrics, validating that the proposed top- k method captures deeper intrinsic object-CAD relationships beyond explicit labels.

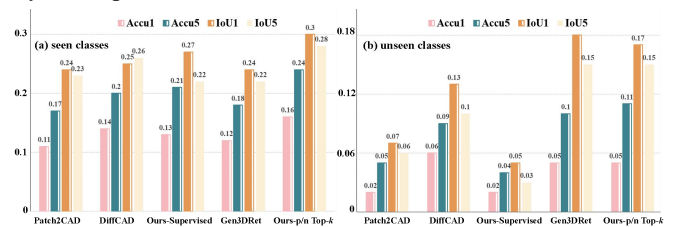


Fig. 7. Model retrieval capability on seen and unseen objects.

Fig. 8 further illustrates **W-MR**'s performance on seen and unseen objects of different categories across multiple viewpoints ($K=5$). Obviously, for seen categories, retrieved CAD models closely match ground-truth geometry and scale. For unseen objects, top-5 retrieved models accurately reflect class characteristics. Although minor geometric mismatches occasionally occur (e.g., chair backrest/armrests shapes in the second instance), top-5 retrieved models still capture overall geometric and semantic characteristics of the target object. Moreover, even under occlusion (e.g., the first instance), the retrieval result remains reasonable, confirming the compatibility of voxel-based feature embedding with **W-MR**.

Given that chairs in the weakly supervised dataset are numerous and difficult to discriminate at a fine-grained level, we evaluate Top-1 retrieval on unseen chair categories using rendered models outside the weakly supervised training set (Fig. 9). Without any explicit annotation, **W-MR** returns Top-1 results that respect chair sub-types, matching key geometric attributes (backrest shape, armrests, base design). This corroborates Fig. 8, indicating that the voxel-based joint

embedding with a differentiable Top- k objective effectively captures “similar but not identical” geometric differences.

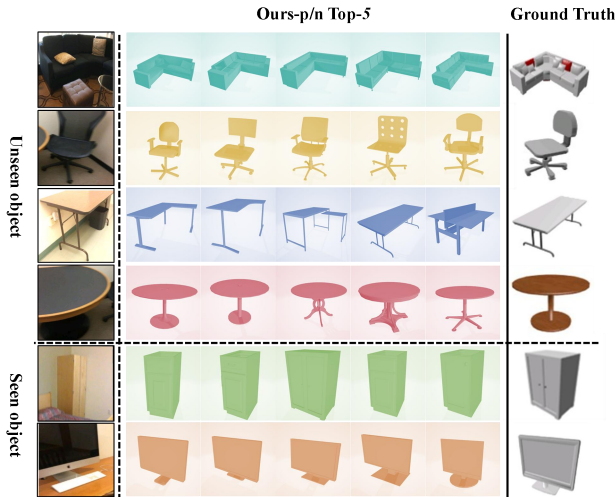


Fig. 8. Qualitative results on some seen and unseen objects.

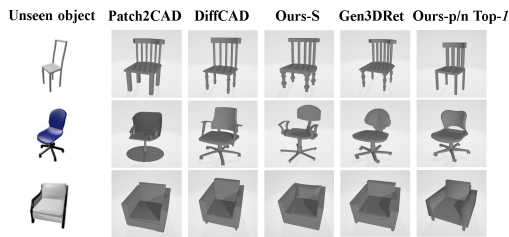


Fig. 9. Qualitative results for unseen chair categories. **Ours-S** denotes **Ours-supervised**.

C. Performance in Real-World Agents

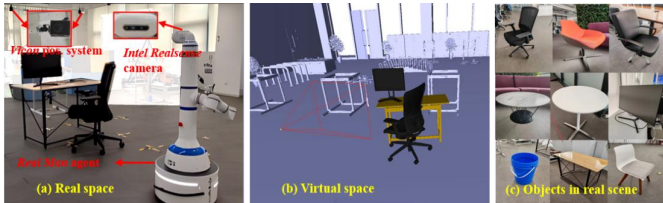


Fig. 10. Real robot navigation experiment.

1) Experimental Settings In this section, we conduct a real-robot indoor navigation experiment (Fig. 10), to evaluate the applicability of the proposed method on physical agents. In this experiment, the mobile agent *RealMan*, equipped with motion-control and task-execution capabilities, autonomously moves from a specified start point to a target point while perceiving environmental changes in real time via an *Intel RealSense D435* visual sensor. The proposed method (**W-MR**-based *VirEqPhy*) takes scene images as input and continuously updates a pre-established virtual space generated in Open3D. The ground-truth camera pose is provided by a *Vicon* motion-capture system with 21 high-precision infrared cameras installed above and around the experimental area.

2) Results Fig. 11 illustrates virtual scene updates under static object layouts, where the green dots represent the camera trajectory. Our method successfully maintains geometric consistency between the virtual and physical scenes: object positions, scale and orientations closely match their

real-world counterparts without noticeable misalignment or confusion. Even for uncommon scene layouts (e.g., a *chair* leaning against or placed on top of a *table*), the system robustly updates the virtual scene.

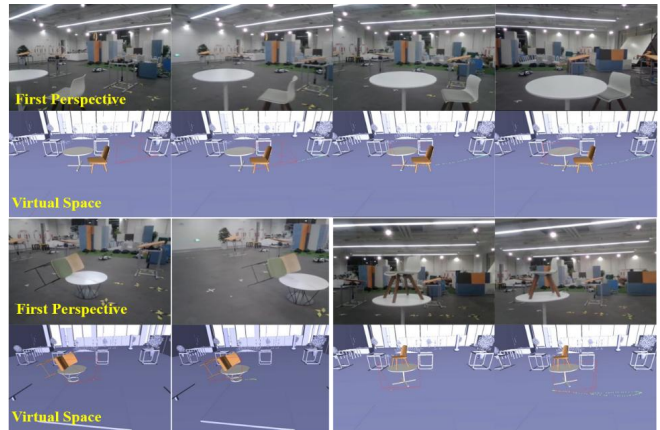


Fig. 11. Virtual space updates in static scene.

For more experimental analysis, please refer to *APPENDIX* section in <https://forbiddenname.github.io/VirEqPhyDT>.

VII. CONCLUSION AND FUTURE WORK

This study introduces *VirEqPhy*, a novel geometric update method that establishes real-time equivalence between virtual and physical spaces of digital twin. Its primary goal is to provide embodied agents with a virtual learning environment mirroring real-world conditions. *VirEqPhy* integrates a unified framework for feature extraction, CAD model retrieval, and object pose estimation within an end-to-end multi-task cascade network, augmented by a weakly supervised module for model retrieval extension. Experiments demonstrate that 1) the approach delivers state-of-the-art high-threshold retrieval and second-best mean retrieval, achieves the best class-level and second-best instance-level alignment, and traces a clear accuracy-latency frontier; 2) under weak supervision, the proposed **W-MR** surpasses retrieval baselines overall, forms tighter intra- and inter-class clusters, and markedly boosts Top-5 on unseen classes while keeping Top-1 performance near-optimal; and 3) the system is feasible for real-world agent navigation across diverse scene layouts.

Building on these technical contributions, *VirEqPhy* offers practical benefits for robotics researchers and practitioners, with potential applications in smart warehouses, indoor service robots, and security inspection. For example, dynamically updated digital twins allow robots to optimize path planning and obstacle avoidance in warehouse environments, while in home service scenarios the system tracks object additions, removals, and pose changes to support skill learning. Moreover, its lightweight design—compatible with edge devices like the NVIDIA Jetson series and not reliant on expensive 3D sensors or fully annotated datasets—significantly lowers deployment barriers. In addition, leveraging domain-specific CAD libraries together with federated learning for cross-scene knowledge transfer can further accelerate real-world adoption.

Despite these advancements, several limitations remain. First, for objects without matching CAD models in the database, the current method relies on query-based retrieval of the most similar model. Future research could investigate general geometric representation estimation (e.g., neural implicit fields like NeRF or SDF) to enable virtual space updates for unseen objects without CAD models. Second, with advancements in visual sensors, future efforts will incorporate multi-modal inputs (e.g., RGB-D or event cameras) to enhance robustness and accuracy in dynamic environments. Finally, this work primarily targets large rigid objects, with limited attention to small items or non-rigid objects (e.g., deformable clothes). In fact, we have qualitatively evaluated the proposed method on small objects in a multi-agent restaurant-service scenario, and our prior work has explored a real-time deformation rendering algorithm for virtual objects based on physical properties, achieving promising results [11]. Future efforts will integrate these works to broaden the applicability of the current system in more challenging scenarios.

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